Special Relativity

Labix

June 16, 2026

Contents

1 Mechanics in Special Relativity **3**
1.1 Minkowski Spacetime 3
1.2 Relativistic Kinematics 5
1.3 Relativistic Dynamics 7
1.4 Coordinate Lagrangian Formulation 8
1.5 Covariant Lagrangian Formulation 8

2 Electromagnetism in Special Relativity **9**

1 Mechanics in Special Relativity

1.1 Minkowski Spacetime

Einstein's two postulates.
Postulate: Homogeneity of space and time and isotropy.

From the above, conclude that spacetime is Minkowski.

Definition 1.1.1 (Minkowski Spacetime)
Define the Minkowski spacetime by the following data.

- A four dimensional affine space $(M, V, +)$.
- A non-degenerate symmetric bilinear form $\eta : V \times V \to \mathbb{R}$ of signature $(3, 1)$.

By Sylvester's law of inertia, we are guaranteed global Lorentz orthonormal bases $E = \{v_0, \dots, v_3\}$. These are the bases for which

$$[\eta]_{E,E} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Typically, we think of η has having the matrix

$$\begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where c is the speed of time. But the geometry does not differ other than by a scalar because it only depends on the signature.

We recall some facts about affine spaces that makes them a smooth manifold.

- For each $p \in M$, the map $\phi_p : M \to V$ given by $\phi_p(x) = x - p$ does not define a global chart on M. We need to choose a basis for V to get an isomorphism $V \cong \mathbb{R}^4$ for the global chart.
- For each $p \in M$, the map $\alpha_p : T_pM \to V$ given by $[\gamma] \mapsto \frac{d}{dt}\bigg|_{t=0}(\gamma(t) - p)$ is a canonical isomorphism. This means that no connection is needed to compare the fibers of the tangent bundle. The structure of the affine space means inherently that M is flat.

We also recall some definitions of Lorentzian geometry. Let V be a vector space of dimension 4. Let η be a non-degenerate symmetric bilinear form on V of signature $(3, 1)$. Let $v = (t, v_1, v_2, v_3) \in \mathbb{R}^4$.

- We say that v is space-like if $|v|_\eta > 0$.
- We say that v is time-like if $|v|_\eta \in \mathbb{C} \setminus \mathbb{R}$.
- We say that v is light-like if $|v|_\eta = 0$.

Definition 1.1.2 (Time-Orientation)
Let $(M, V, +, \eta)$ be Minkowski spacetime. A time orientation of M is a choice of connected component of the set of time-like vectors $\{v \in V \mid |v|_\eta \in \mathbb{C} \setminus \mathbb{R}\}$.

Definition 1.1.3 (Future Pointing Vectors)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M. Let $v \in V$ be a time-like vector. We say that v is future pointing if v lies in the connected component defining the time orientation.

Definition 1.1.4 (The Minkowski Metric)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Define the Minkowski metric $\eta \in \mathcal{C}^\infty(T^*M \otimes T^*M)$ as

follows. For each $p \in M$, define

$$\eta_p(v, w) = \eta(\alpha_p(v), \alpha_p(w))$$

where α_p is the isomorphism $\alpha_p : V \xrightarrow{\cong} T_pM$.

The Lorentzian properties on V translate canonically to T_pM via α_p. This gives M the structure of a pseudo-Riemannian manifold.

Proposition 1.1.5
Let $(M, V, +, \eta)$ be Minkowski spacetime. Then (M, η) is a pseudo-Riemannian manifold with metric signature $(1, 3)$.

For a curve $\gamma : \mathbb{R} \to M$ in Minkowski space time, the vectors $\gamma'(t) \in T_pM$ is either space-like, time-like or light-like. Solving the equation $|x|_\eta = 0$ gives a 3-dimensional cone in $\mathbb{R}^4$.

- If $\gamma'(t)$ lies on the light cone, then $\gamma(t)$ is travelling at the speed of light.
- If $\gamma'(t)$ lies inside the light cone (time-like), then this is physically realizable as an object travelling through space time.
- If $\gamma'(t)$ lies outside the light cone (space-like), then it is physically impossible.

Definition 1.1.6 (Inertial Frame of Reference)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Fix a time orientation of M. An inertial frame of reference consists of the following data.
- A choice of the origin $p \in M$.
- A Lorentz orthonormal basis $e_0, \dots, e_3$ of V such that e_0 is future pointing.

Let $E = \{e_0, \dots, e_3\}$. Let $\phi_E : V \xrightarrow{\cong} \mathbb{R}^4$ be the induced isomorphism. Then $(M, \phi_E \circ \phi_p)$ is a chart on M.

Definition 1.1.7 (The Poincare Group)
Let V be a vector space of dimension 4. Define the Poincare group to be

$$P = O(1, 3) \ltimes V$$

Proposition 1.1.8
Let $(M, V, +, \eta)$ be Minkowski spacetime. Then there is an isomorphism

$$\text{Isom}(M, \eta) \cong O(1, 3) \ltimes V$$

Proposition 1.1.9
Let $(M, V, +, \eta)$ be Minkowski spacetime. Then the Poincare group is the largest subgroup of Aff(V) that maps inertial frames to inertial frames. Also it acts transitively on inertial frames.

Definition 1.1.10 (Translations, Boosts and Rotations)
Let $T \in O(1, 3) \ltimes \mathbb{R}^4$.
- We say that T is a boost if T is of the form

$$T(t, x) = \left(\frac{t - v \cdot x}{\sqrt{1 - v^2}}, \frac{\frac{\langle x, v \rangle}{|v|} - |v|t}{\sqrt{1 - v^2}} + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right) \right)$$

  for some $v \in \mathbb{R}^3$ such that $|v| < 1$, and $x = \frac{\langle x, v \rangle}{|v|^2} v + \left(x - \frac{\langle x, v \rangle}{|v|^2} v \right)$ is the orthogonal decomposition.

- We say that T is a translation if T is of the form
$$T(t,x) = (t+t_0, x+x_0)$$
for some $x_0 \in \mathbb{R}^3$ and $t_0 \in \mathbb{R}$.
- We say that T is a rotation if T is of the form
$$T(t,x) = (t, Gx)$$
for some $G \in SO(3) \leq SO^+(1,3)$.

Boosts are different in Minkowski spacetime because we assume $c=1$ and no velocity can reach $c=1$. So $|v|<1$.

In the Galilean spacetime these three types decomposes the Galilean group. In Minkowski spacetime, boosts and rotations are given by the Cartan decomposition $SO^+(1,3) \cong \{\text{Boosts}\} \times SO(3)$ as a smooth manifold (not as a group). Combining the translations give the restricted Poincare group, which is not the full Poincare group.

Definition 1.1.11 (The Restricted Poincare Group)
Define the restricted Poincare group to be
$$P^+ = \mathbb{R}^4 \rtimes SO^+(1,3)$$

Proposition 1.1.12
There is an isomorphism
$$P \cong P^+ \ltimes_\phi (\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z})$$
where $\phi : \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to \text{Aut}(P^+)$ is given by:

1.2 Relativistic Kinematics

Definition 1.2.1 (World Lines)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a curve. We say that r is a world line if $r'(t)$ is future pointing for all $t \in \mathbb{R}$.

Physically: time is coordinate dependent, proper time is the absolute notion that everyone agrees on.

Definition 1.2.2 (Proper Time)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a world line. Let $t_0 \in I$. Define the proper time of r to be
$$\tau(t) = \int_{t_0}^t \sqrt{-\eta(r'(s), r'(s))}\, ds$$
for $t \in I$ and $t > t_0$.

Heuristically, we have
$$d\tau = \sqrt{-\eta(r'(s), r'(s))}dt$$
The choice of coordinate frame changes how η and dt is presented, but $d\tau$ stays the same. If r is light-like instead, notice that $d\tau = 0$. Photons traveling in vacuum does not experience time in the same sense.

Time Dilation: Choose an inertial frame. Then r is parametrized as $r(t) = (ct, x(t), y(t), z(t))$. Let $v = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right)$. Then we have $d\tau = \sqrt{1-|v|^2}dt$. So time experienced by an observer is proportional to proper time by a factor depending on the velocity of the observer.

The Twin Paradox: Among all world lines r passing through p and q, the geodesic one (the straight line) has the largest proper time. The twin paradox says that the twin taking the bendy road with acceleration ages faster than the other twin whose trajectory is straight. Mathematically, $r(\tau)$ is the unit speed parameterization of r.

Definition 1.2.3 (Unit Parametrization)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a world line. Define the unit parametrization of r by
$$\widehat{r}(s) = (r \circ \tau^{-1})(s)$$

Definition 1.2.4 (Lorentz Factor)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a world line. Define the Lorentz factor of r to be
$$\gamma = \frac{1}{\sqrt{-\eta(r', r')}}$$

Definition 1.2.5 (Four-Velocity and Four-Acceleration)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a world line.
- Define the four-velocity of r by
$$u(s_0) = \frac{d\widehat{r}}{ds}\bigg|_{s=s_0} = \gamma r'(s_0)$$
- Define the four-acceleration of r by
$$a(s_0) = \frac{du}{ds}\bigg|_{s=s_0}$$

Because tangent spaces are canonically identified in Minkowski spacetime, we think of $a(s_0)$ as an element of the tangent space $T_{\gamma(s_0)}M$.

Proposition 1.2.6
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $r : I \subseteq \mathbb{R} \to M$ be a world line. Then the following are true.
- $\eta(u, u) = -1$.
- $\eta(u, a) = 0$.

In Galilean spacetime, the three dimensional slices of the world along time is orthogonal to the flow of time. In Minkowski spacetime, the same is true, but time flows differently according to each observer (inertial frames).

Definition 1.2.7 (Simultaneity Hypersurface)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let $v \in V$ be a future pointing time-like vector. Let $p \in M$. Define the simultaneity hypersurface through p relative to v by the space
$$\Sigma_p(v) = \{q \in M \mid \eta(q - p, v) = 0\}$$

A key point is that $\eta|_{\Sigma_p(v)}$ is positive definite, which means that it becomes an inner product on the three dimensional slice. Another key point: for the worldline r_1, r_2 of two observers at the same point p, their future pointing time-like vectors r_1' and r_2' are generally different, which gives a different three dimensional slice.

Length contraction: Suppose an metal rod exists in spacetime. The endpoints are given by two world lines r_1, r_2 such that $r_2 = q + r_1$ for some $q \in V$ and $\eta(r_2 - r_1, v)$ where here we choose an inertial frame to fix a future pointing vector v. Then define the length of the rod according to this observer to be the quantity
$$L = \sqrt{\eta(q, q)}$$

Then???

If a basis E is chosen for V such that

$$[\eta]_{E,E}\begin{pmatrix} -c^2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then unit parameterization would instead be

$$\tau(t) = \frac{1}{c}\int_{t_0}^{t} \sqrt{-\eta(r'(s), r'(s))}\, ds$$

and we would have $\eta(u, u) = -c^2$. Time dilation is displayed as $d\tau = \sqrt{1 - |v|^2/c^2}dt$

1.3 Relativistic Dynamics

In classical mechanics, Galilean force laws are of the form

$$m\frac{d^2r}{dt^2} = F$$

where F is invariant under Galilean transformations.

Definition 1.3.1 (Minkowski Force)
Let $(M, V, +, \eta)$ be Minkowski spacetime. A force law consists of the following data.

- A world line $r : I \subseteq \mathbb{R} \to M$.
- A constant $m \in \mathbb{R}$ called rest mass.
- A smooth vector field $F : I \to TM$ called a force such that

$$ma = F$$

Definition 1.3.2 (Four Momentum)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Define the four momentum to be

$$p = mu$$

Proposition 1.3.3
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Then the following are true.

- $\eta(p, p) = m^2$.
- $\eta(p, F) = 0$.

For the rest of the section here we assume that η is given specifically by the matrix that incorporates the speed of light c. Assume that ordinary Newtonian force law in three dimensions hold. In other words we assume that

$$\frac{d}{dt}\frac{mr'(t)}{\sqrt{1 - |r'(t)|^2/c^2}} = \mathbf{F}$$

for some Newtonian force law F (determined for example experimentally). Using $\eta(p, F) = 0$, we deduce that $F = \left(\frac{\gamma}{c}\mathbf{F} \cdot r'(t), \gamma\mathbf{F}\right)$. Comparing first components of a force law gives us

$$\frac{d}{dt}\frac{mc^2}{\sqrt{1 - |r'(t)|^2/c^2}} = \mathbf{F} \cdot r'(t)$$

Since integrating $\mathbf{F} \cdot r'(t)$ gives work done, we must make the following definition.

Definition 1.3.4 (Relativistic Energy)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Define the relativistic energy to be

$$E = \gamma mc^2$$

Using the binomial theorem, we see that

$$E = mc^2 + \frac{1}{2}m|r'(t)|^2 + \cdots$$

The higher terms contain velocity $|r'(t)|$. So the first term recovers Einstein's famous equation.

Proposition 1.3.5
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. Then we have

$$E^2 = p^2c^2 + m^2c^4$$

1.4 Coordinate Lagrangian Formulation

We first consider a single free particle in Minkowski spacetime. Ideally we want the Lagrangian to be invariant under relativistic quantities. So assume that

$$L = \alpha\gamma$$

for some $\alpha \in \mathbb{R}$. Here the Lorentz factor γ is thought to be the simplest relativistic invariant quantity. Then variational principle solution to the action $S = \int_a^b L\, dt$ is precisely the solution to the usual Euler Lagrange equation in classical mechanics. When there are other forces acting on the particle, we assume that it interacts on the Lagrangian simply.

Definition 1.4.1 (The Coordinate Lagrangian)
Let $(M, V, +, \eta)$ be Minkowski spacetime. Let (r, m, F) be a force law. A Lagrangian is a function $L : TM \times \mathbb{R} \to \mathbb{R}$ of the form

$$L = -\gamma mc^2 - V$$

for some function $V : TM \times \mathbb{R} \to \mathbb{R}$ called the potential.

The problem here is that V is in general not Lorentz invariant and so L is in general not Lorentz invariant.

Pros: Motions of equations are accurate and easy enough.
Cons: Not covariant inherently, although in the usual cases they are provable to be covariant.

1.5 Covariant Lagrangian Formulation

Pros: It is covariant. Cons: There are difficulties generalizing to N-bodys.

2 Electromagnetism in Special Relativity